1. Introduction to AI

1. What is AI?

AI is the science of developing techniques to generate intelligence artificially — building machines that can *think like humans* (reason, learn, perceive, and decide).

It is often described as a **disruptive technology**: a technology that doesn't just improve existing tools but fundamentally reshapes industries and the way work is done.

Types of AI Based on Capabilities

- **ANI (Artificial Narrow Intelligence):** Also known as **"Weak AI."** This AI is highly specialized and trained to excel at one specific task (e.g., face recognition, chess engines, language translation). It operates under strict constraints and cannot apply its intelligence to any problem outside its domain.
- **AGI (Artificial General Intelligence):** Also known as **"Strong AI"** or human-level AI. This represents a hypothetical machine that possesses the ability to understand, learn, reason, and apply knowledge across any intellectual task that a human being can do.
- **ASI (Artificial Superintelligence):** A futuristic, hypothetical form of AI that surpasses human intelligence across all domains, including scientific creativity, emotional intelligence, general wisdom, and social skills.

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2. A Bit of History — When Did AI Start?

The field was formally named in **1956** at the **Dartmouth Summer Research Project on Artificial Intelligence** (the Dartmouth Conference). Key founding figures include:

- **John McCarthy** — coined the term *"Artificial Intelligence"* and later created the **LISP** language.
- **Marvin Minsky** — co-founder of the field and of the MIT AI Lab.

(Other founders present in this era included Nathaniel Rochester and Claude Shannon.)

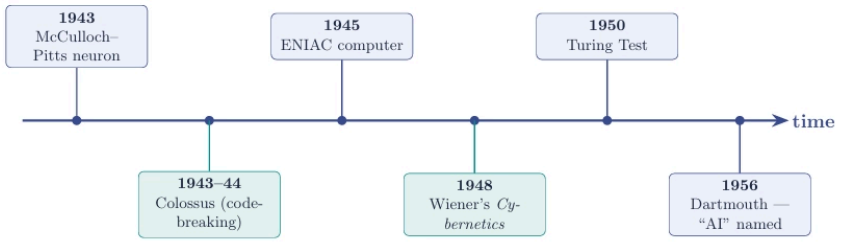
How World War II acted as a catalyst for AI

Although AI was *named* in 1956, the intellectual and technological groundwork was laid during **World War II (1939–1945)**. The war turned abstract ideas about "thinking machines" into urgent, funded, practical problems:

- **Codebreaking and Alan Turing.** At Britain's Bletchley Park, Turing helped design the electromechanical **Bombe** to break the German **Enigma** cipher. This showed that a machine could carry out tasks that previously seemed to require human reasoning — a direct conceptual seed for AI.
- **Birth of electronic computers.** The war drove the creation of the first programmable electronic machines: **Colossus** (1943–44, built to break the Lorenz cipher) and, just after the war, **ENIAC** (1945, built to compute artillery firing tables). AI needs computers to run on — and the war is what produced them.

- **Cybernetics and feedback control. Norbert Wiener** worked on automatic anti-aircraft fire-control systems that had to *predict* a plane's future position. This work on feedback and control led to his 1948 book *Cybernetics*, a foundation for thinking about machines that sense, adjust, and act — the same loop that defines intelligent agents today.
- **The artificial neuron.** In **1943**, McCulloch and Pitts published a model showing how networks of simple neuron-like units could compute logical functions — the ancestor of today's **neural networks**.
- **Turing's 1950 paper.** Building on his wartime experience, Turing published *"Computing Machinery and Intelligence,"* proposing the **Turing Test** ("Can machines think?").
- **The Turing Test -** Proposed by Alan Turing in his landmark 1950 paper, this behavioral test evaluates whether a machine can exhibit intelligent behavior indistinguishable from a human.
  • **The Setup:** A human judge carries out a text-based, natural language conversation with two hidden entities: one human and one machine.
  • **The Criterion:** If the judge cannot reliably distinguish the machine from the human after a period of conversation, the machine passes the test.
- The term **"Machine Learning"** was officially coined by **Arthur Samuel** in 1959 during his tenure at IBM. He famously defined it as giving computers the ability to learn without being explicitly programmed, demonstrating this by building a checkers program that learned to play better than its creator.

In short, WWII supplied AI with three things it could not have started without: **working computers, funding and urgency, and a generation of scientists** who had personally seen machines do "intelligent" work. The 1956 Dartmouth Conference then gave the field its name.



From wartime computing to the founding of AI.

3. Two Major Approaches to Building AI

Approach	Idea	Examples / Tools
GOFAI ("Good Old-Fashioned AI")	Encode human knowledge as explicit **rules and logic**; the machine reasons by following them. Also called the **symbolic / rule-based** approach.	**LISP**, **Prolog**, expert systems
Data-Driven Approach	Don't hand-code rules — let the machine **learn patterns from data**.	Machine learning, deep learning, transformers

The history of AI is largely a shift **from the rule-based GOFAI approach toward the data-driven approach** that dominates today.

Why the Data-Driven Approach is Powerful Today

The historical paradigm shift away from rule-based GOFAI toward modern data-driven approaches succeeded due to two primary modern catalysts:
1. **Availability of Data (The Big Data Explosion):** Early AI lacked the sheer amount of training data required to find meaningful patterns. Today's hyper-connected world continuously generates vast reservoirs of information.
2. **Massive Computing Power:** Modern deep learning relies heavily on intense matrix mathematical operations. The rise of specialized hardware (GPUs and TPUs) along with scalable cloud computing infrastructure made processing these massive computations feasible.

What is Big Data? (Introduced in 2001)

The data explosion that fuels modern machine learning algorithms is known as **Big Data**. Characterized by analyst Doug Laney in 2001, it is defined by the **3 Vs**:
• **Volume:** The sheer, massive scale and quantity of data generated and stored every single second.
• **Velocity:** The incredible speed at which new data is created, streamed, and needs to be processed in real time.
• **Variety:** The structural diversity of data, encompassing structured data (databases), semi-structured data (XML, logs), and unstructured data (text, video, audio formats).

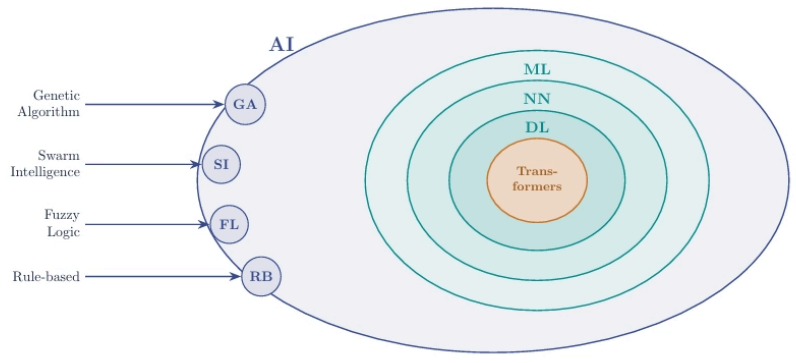
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4. The AI Landscape

AI is a broad umbrella containing many techniques. Some are classical / standalone methods; others are nested inside one another.

Classical & complementary techniques (under the AI umbrella):

- **GA — Genetic Algorithms** (optimization inspired by evolution)
- **SI — Swarm Intelligence** (behaviour inspired by ants, birds, etc.)
- **FL — Fuzzy Logic** (reasoning with "degrees of truth" rather than strict true/false)
- **RB — Rule-Based systems** (the GOFAI style above)



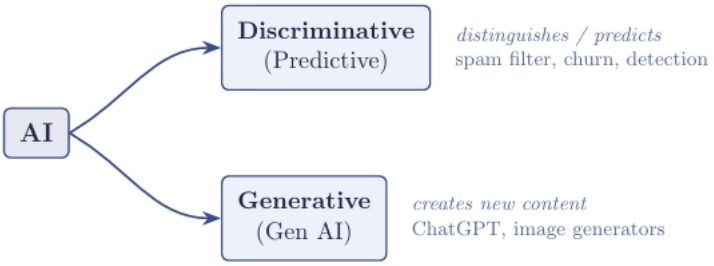
GA Genetic Algorithms • **SI** Swarm Intelligence • **FL** Fuzzy Logic • **RB** Rule-Based • **ML** Machine Learning • **NN** Neural Networks • **DL** Deep Learning

- **ML** — algorithms that learn from data.
- **NN** — models loosely inspired by the brain.
- **DL** — many-layered neural networks.

- **Transformers** — the architecture behind modern large language models (e.g., ChatGPT).

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5. Discriminative vs. Generative AI

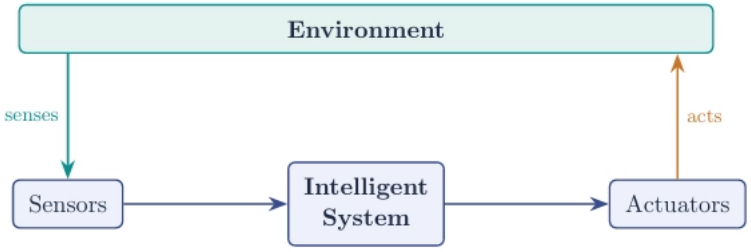


- **Discriminative / Predictive AI** — *distinguishes* between things: classification, prediction, detection (e.g., "is this email spam?", "will this customer churn?").
  - **Three Core Types of Discriminative Tasks**
    - **Classification:** Used when the model predicts a discrete **class or category** label (e.g., predicting whether an email is "Spam" or "Not Spam", or identifying an image as a "Cat" or "Dog").
    - **Regression:** Used when the model predicts a continuous **numerical value** (e.g., predicting the exact price of a house, a person's age, or future revenue metrics).
    - **Time Series Forecasting:** Used when the model predicts a **sequential or temporal value** based on historical, time-ordered data points (e.g., predicting tomorrow's weather patterns, hourly server load, or stock market fluctuations over the next week).
- **Generative AI (Gen AI)** — *creates* new content: text, images, audio, code (e.g., ChatGPT, image generators).

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6. Intelligent Agents — How an AI System Interacts with the World

An intelligent system operates in a continuous **sense** → **think** → **act** loop with its environment:



- **Sensors** — let the system *sense* / perceive the environment (cameras, microphones, inputs).
- **Intelligent System** — processes the input and decides what to do.
- **Actuators** — let the system *act* on the environment (motors, outputs, responses).

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7. Embodied vs Disembodied AI

Type	Meaning	Example
Embodied AI	AI that has a **physical body** and acts in the real world through sensors and actuators.	**Self-driving car**
Disembodied AI	AI that exists **purely as software**, with no physical form.	**ChatGPT**